DSAA 2026 · SPECIAL SESSION- LLFM



S3 Labs – IIT Bhilai

IEEE DSAA2026

ZoneGate

Deployment-Readiness Audit of Vision-Language Models for Industrial Safety Monitoring

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Problem Statement

1,000+
fixed CCTV cameras across the plant

12 + 50
full-time + part-time safety officers

~20
feeds watched by each officer at once

20–30 m
overhead views through dust, steam, glare

The deployment question

Not: how does a VLM score on average?

But: in which production zones can it be trusted to assist review, and where must a human take over?

Constraint: *Each zone has its own rules, activities and visibility, so reliability is a per-zone property.*

Related Work

VLMs for industrial safety

Hazard detection, PPE compliance, MonitorVLM, iSafetyBench, IndustryEQA.

Gap: Report held-out scores, not whether a model can be acted on in a given work area.

Deployment-readiness studies

CRISP-ML(Q), hidden technical debt, industrial case studies.

Gap: Judge readiness for the whole system, not area by area.

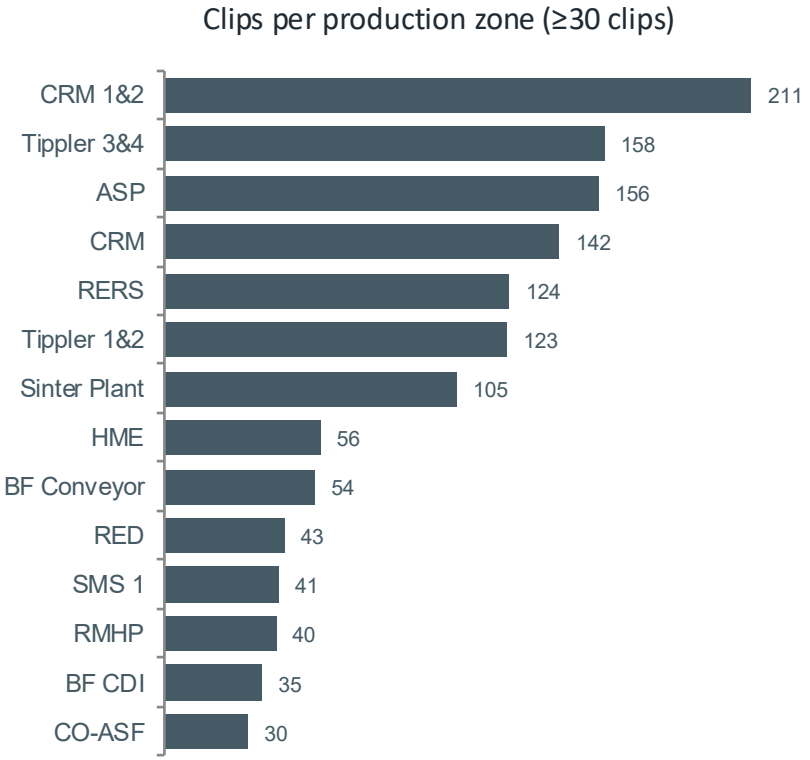
Selective prediction & deferral

Reject option, SelectiveNet, learning to defer.

Gap: Operate per instance and assume a usable confidence signal.

ZoneGate: selective prediction applied at the production-zone level, on live plant video, returning a ROUTE / ABSTAIN decision per zone.

Pilot Deployment Setup



1,345 clips · 64 cameras · 67 camera-days

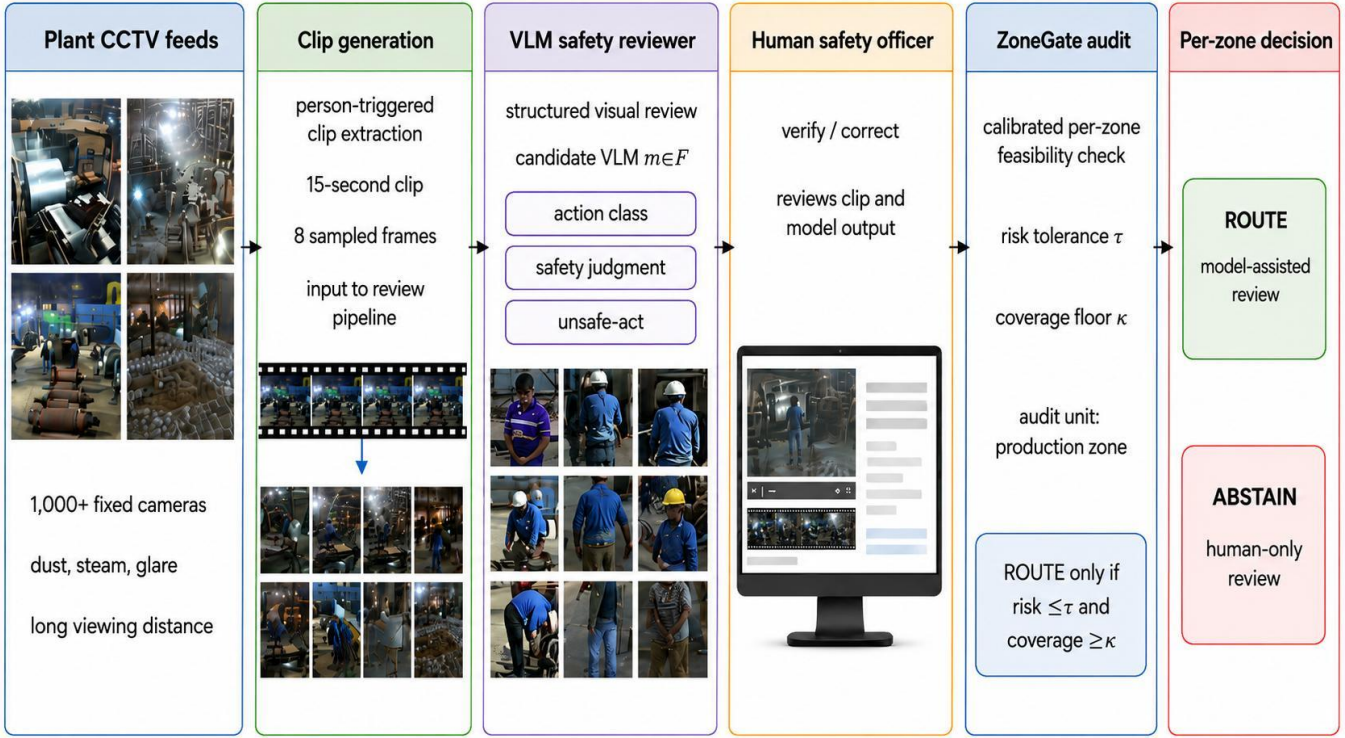
15 s person-triggered clips (YOLOv8n), 8 frames each (Inference is image based)

2 arms 795 model-assisted vs 550 direct review, randomised within zone

AC1 0.77 safety-flag agreement (action label IAA is 0.71)

Deployed model: Qwen3-VL-235B, structured JSON (action, safety judgment, unsafe act). Released as SteelBench on Hugging Face.

ZoneGate workflow



ZoneGate audit gives us trail on why the specific model routes or abstains a clip and handovers to a human review

METHOD

ZoneGate Method(1)

1 Retained set

$$R_z(t) = \{ i \in z : c_i \geq t \}$$

clips where self-reported confidence clears threshold t

2 Coverage

$$\varphi_z(t) = |R_z(t)| \,/\, |z|$$

share of the zone the model decides

3 Selective risk

$$r_z(t) = \frac{1}{|R_z(t)|} \sum_{i \in R_z(t)} \mathbb{1}[\hat{y}_i \neq y_i] .$$

error rate on the retained clips

4 Deployable coverage

$$DC_z(\tau) = \max \{ \varphi_z(t) : r_z(t) \leq \tau \}$$

largest share decidable at risk ≤ τ (0 if none)

ZoneGate Method(2)

1 · Calibrate elsewhere

$$t_z^\star = \arg\max_{t\in[0,1]} \phi_C(t) \quad \text{s.t.} \quad r_C(t) \le \tau,$$

with C = all zones except z (leave-one-zone-out)

2 · Feasible?

If no threshold meets τ on the calibration pool, t* = ⊥

3 · Enough coverage?

Apply t* unchanged to zone z: is $\phi_z(t^*) \ge \kappa$? Uses only confidences, no labels from z

ABSTAIN if t* = ⊥

ROUTE if $\phi_z(t^*) \ge \kappa$

Operating points

τ = 0.20
strict safety, main criterion

τ = 0.30
relaxed, how close to routable

τ = 0.40
calibration-transfer diagnostic only

κ = 0.50
must retain half the zone

A fixed rule helps the safety officer trace exactly why a zone abstains.

Safety Profile of Model Assisted Arm

	Expert label → Violation	Safe
Flag	**341** true positives	**264** false alarms
No flag	**46** missed violations	**144** true negatives

795 model-assisted clips, Qwen3-VL-235B

0.88

recall: few real violations are missed

0.56

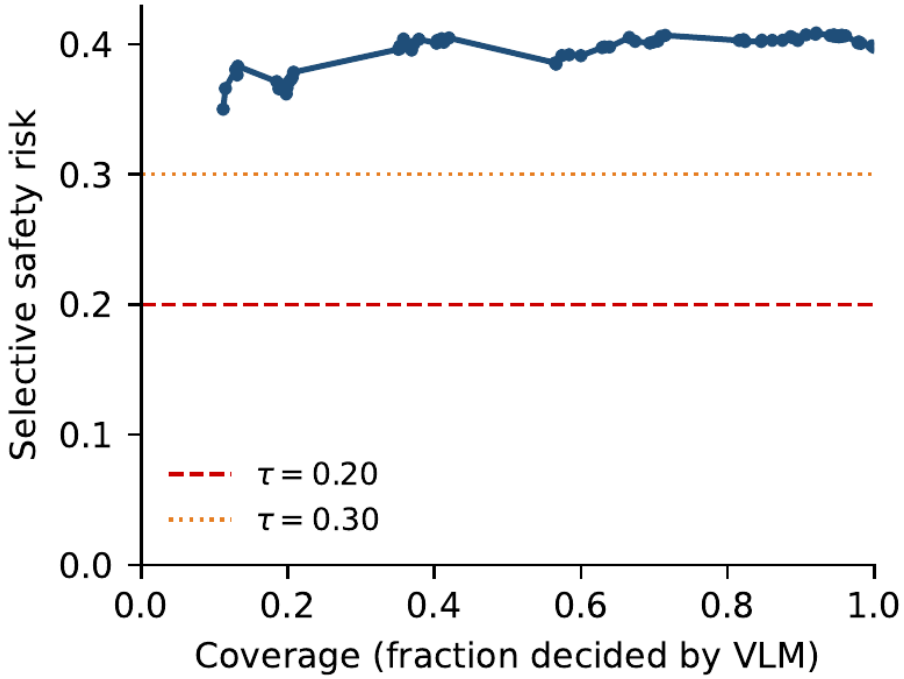
precision: almost half of alerts are false

>85%

of all errors are false alarms on safe scenes

Per-zone recall ranges from 0.64 (SMS 1) to 1.00 (RERS). The recall–alert frontier never reaches a point where false alarms fall away while recall holds.

Selective Risk Analysis



~0.39

selective risk across the whole coverage range

0.37

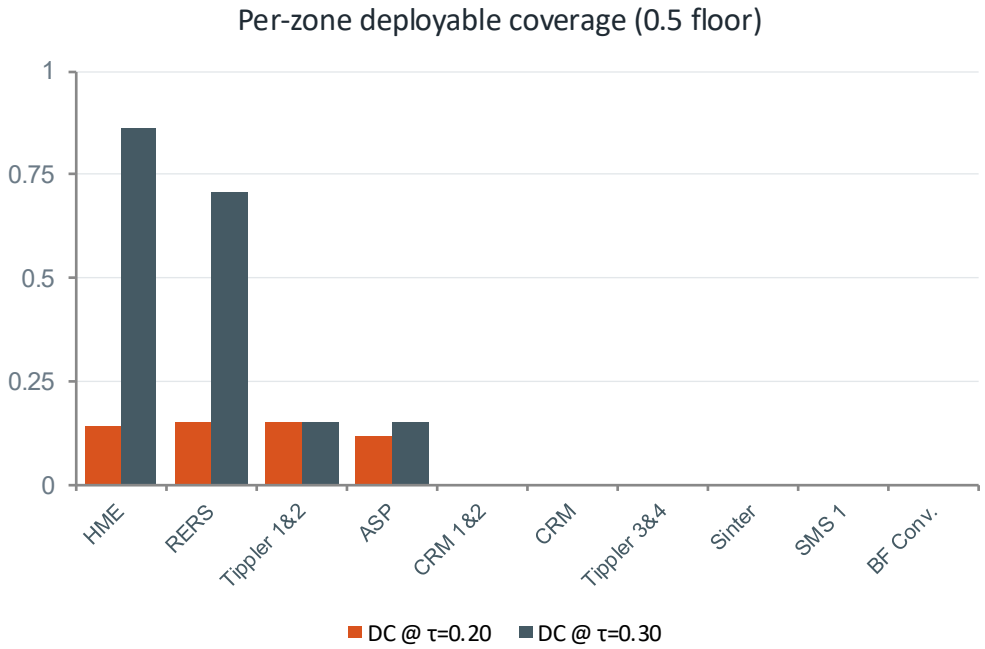
only on the most-confident tenth of clips

DC(0.20) = 0

bootstrap 95% CI [0.00, 0.00]

Confidence can track scene legibility but not the safety call

Impact of Threshold and Model Selection



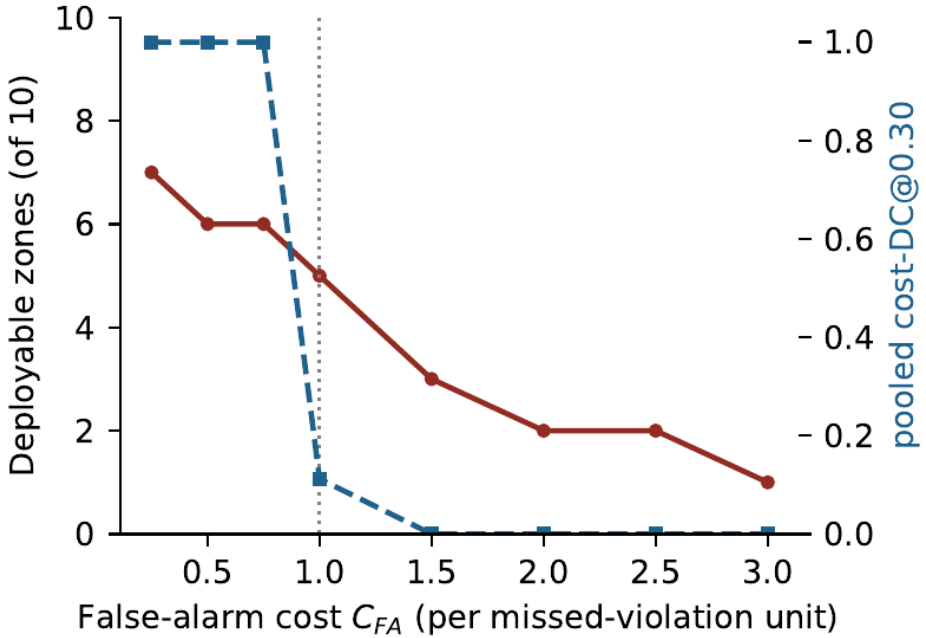
Per-zone (oracle) DC. Only HME and RERS pass at 0.30; after leave-one-zone-out calibration, no zone routes at τ ≤ 0.30.

All nine candidate VLMs

Claude Opus 4.7	**0.00**
GPT-5.4	**0.00**
GPT-4o	**0.00**
Gemini 2.5 Pro	**0.00**
Gemini 2.5 Flash	**0.00**
Qwen3.5-122B	**0.00**
Llama-4 Maverick	**0.00**
Gemma-4-31B	**0.00**
Nemotron-12B	**0.00**

DC at τ = 0.20 (best at 0.30: 0.01)

Deployability is set by the price of a false alarm



6 → 1

zones clearing the floor at τ = 0.30 as the false-alarm cost rises

Weighted risk

Violations weighted by severity (s_i + 1); safe clips weighted by C_{FA}. Unweighted risk: only 2 zones clear.

The plant's choice is how much false alerting its team can absorb, not which model scores highest.

CASE STUDY

Sinter Plant Case Study

26 / 29

violations caught (recall 0.90)

24 / 39

safe clips falsely flagged (precision 0.52)

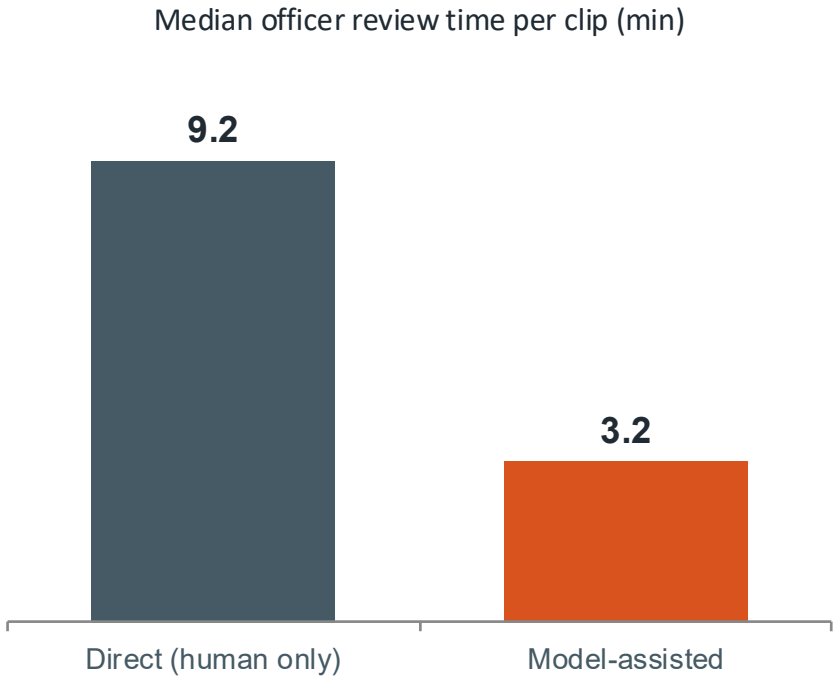
All 3 misses were hi-vis vest cases.

Most frequent false flags

1. Missing dust mask
2. Absent safety goggles
3. Safety shoes
4. High-visibility vest

The equipment was present or not required; experts marked the scene safe. From a ceiling-mounted camera the model cannot confirm PPE, and the conservative prompt makes it flag. No threshold removes these without dropping real violations → ABSTAIN.

Model as a Review Assistant



n = 550 direct vs 795 model-assisted clips

2.9×

faster review (−65%), with the safety call left to the officer

$0.007

inference cost per clip, so the real cost is human review

94.5%

of clips (673 / 712) still deferred at τ = 0.20, about 36 officer-hours over the pilot

Deployment lessons

1 The decision is the zone, not the model

Every candidate leaves DC ≈ 0 at a safe risk, so the gate decides whether to route a zone.

2 The limit is over-cautious reasoning

Survives calibration, all 9 models, and confidence tracks legibility. Errors are PPE the camera cannot confirm.

3 The model is an assistant, not a verifier

9.2 → 3.2 min review time in routed and abstained zones alike.

4 False-alarm tolerance governs deployability

As the false-alarm price rises, routable zones fall from 6 to 1.

What is missing: a safety-specific, calibrated confidence signal, or direct PPE verification, not a bigger general model.

TAKEAWAY

We should report an industrial VLM deployment as a per-zone decision with a clear audit trail to improve the trust

0 / 10

zones route at $\tau = 0.20$

9 / 9

VLMs with DC ≈ 0

2.9×

faster officer review

Data: SteelBench on Hugging Face (huggingface.co/datasets/steelbench/SteelBench), CC BY-NC 4.0

Thank you. Questions?

Per-zone safety on model-decided clips (Table I)

Zone	n	Recall	Prec.	DC@0.20	DC@0.30
CRM 1&2	133	0.72	0.46	0.00	0.00
ASP	96	0.98	0.64	0.12	0.15
CRM	85	0.82	0.48	0.00	0.00
RERS	85	1.00	0.67	0.15	**0.71**
Tippler 3&4	77	0.97	0.51	0.00	0.00
Sinter Plant	68	0.90	0.52	0.00	0.00
Tippler 1&2	67	0.81	0.51	0.15	0.15
HME	36	0.82	0.50	0.14	**0.86**
SMS 1	35	0.64	0.39	0.00	0.00
BF Conveyor	30	0.70	0.78	0.00	0.00

Ten zones with ≥ 30 model-decided clips (712 clips). Green: oracle per-zone DC above the 0.5 floor.

These are per-zone oracle values. The gate uses the leave-one-zone-out threshold, which is infeasible at $\tau \leq 0.30$, so it abstains everywhere.

At $\tau = 0.40$, calibrated thresholds hold target within 6 pts at 0.78 coverage; only 3 / 10 zones clear the floor.

Candidate models and serving cost

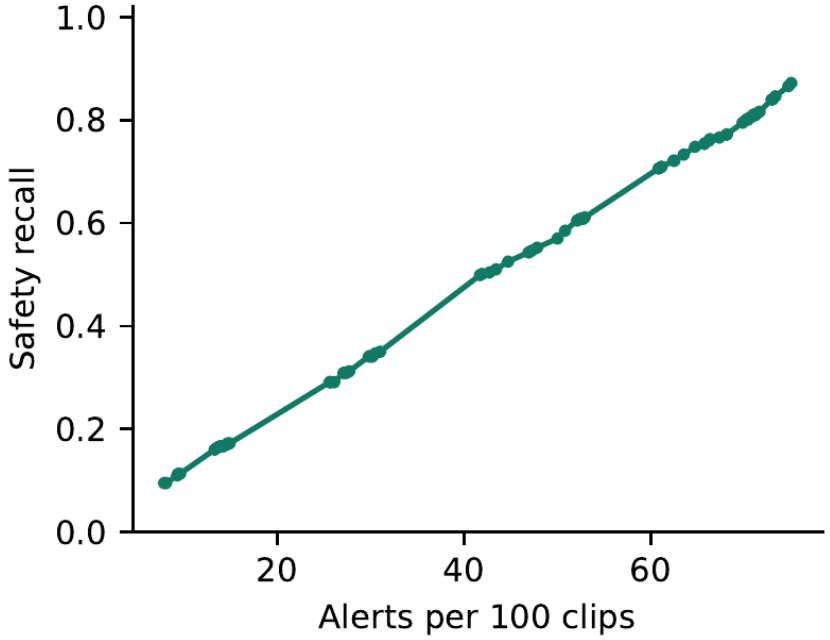
Model	Provider	$/clip	DC@0.20	DC@0.30
Claude Opus 4.7	Anthropic	0.0180	0.00	0.00
GPT-5.4	OpenAI	0.0150	0.00	0.00
GPT-4o	OpenAI	0.0080	0.00	0.00
Gemini 2.5 Pro	Google	0.0060	0.00	0.00
Gemini 2.5 Flash	Google	0.0006	0.00	0.01
Qwen3.5-122B-A10B	Alibaba	0.0015	0.00	0.00
Llama-4 Maverick	Meta	0.0020	0.00	0.00
Gemma-4-31B-it	Google	0.0008	0.00	0.00
Nemotron-Nano-12B-v2-VL	NVIDIA	0.0006	0.00	0.00

Same 8-frame protocol and JSON schema for every model; temperature 0.1.

Costs at mid-2025 public serving rates, used only for cost accounting, never in the routing rule.

Pilot model (officer-facing): Qwen3-VL-235B.

Recall–alert frontier and diagnostic definitions



Alert set: $A_z(t) = \{ i : \hat{y}_i = 1, c_i \geq t \}$

Recall: share of true violations inside $A_z(t)$

Alert load: $100 \cdot |A_z(t)| / |z|$ alerts per 100 clips

Weighted risk: $w_i = s_i + 1$ for violations, c_{FA} for safe clips

Recall rises roughly linearly with alert load: no knee where false alarms drop away while recall holds.

Prompt design and label reliability

System prompt (excerpt)

"You are a trained safety officer reviewing surveillance footage from an integrated steel plant … A human annotator will review your output to verify and correct it, so be detailed, accurate, and conservative."

25 action classes in 6 groups, per-worker PPE state, coordination flag, unsafe act, per-zone rule block.

Conservative by design: keeps misses low, raises false alarms. Relaxing it trades false alarms for missed violations.

0.77 Gwet AC1, unsafe-act flag (86% agreement)

0.71 Cohen κ, action label

0.95 Cohen κ, worker count

1.00 Cohen κ, severity

Scope and limitations

1	**No distribution-free guarantee**	Zones are groups, not exchangeable samples; risk control rests on threshold transfer, which we measure directly.
2	**One plant, one pilot**	1,345 clips, 64 cameras, Nov 2025 to Apr 2026. Other sites may price false alarms differently.
3	**Self-reported confidence**	Used without recalibration; the result implies we need a safety-specific signal, not a better generic one.
4	**Sparse zones**	4 of 14 zones have < 30 model-decided clips and are reported in the supplement only.